

AI Prompt Logs

Date	07 July 2026
Project Name	CaptionAI: AI Powered Social Media Caption Generator
Maximum Marks	2 Marks

AI Prompt Logs:

This log maintains a structured record of interactions with ChatGPT throughout the development of CaptionAI, documenting the objective, the exact prompts used, the resulting output, and how each response was applied to the project.

S.No	Task / Intent	System Prompt & Context	Exact User Prompt	AI Output Summary	Refinements / Iterations (Outcome)
1	Generate project ideas.	No specific system prompt/persona used – general conversational query.	Suggest AI-based web application project ideas suitable for students using Python Flask.	The AI suggested multiple project ideas including an AI-powered social media caption generator. This idea was selected for further development.	Selected the CaptionAI project.
2	Define the problem statement.	No specific system prompt/persona used – general conversational query.	Write a problem statement for an AI-powered social media caption generator that helps users create captions, hashtags, and call-to-action suggestions.	The AI generated a structured problem statement highlighting challenges in creating engaging social media content.	Used the response to prepare the project documentation.
3	Design the application architecture.	No specific system prompt/persona used – general conversational query.	Suggest a system architecture for a Flask-based AI-powered caption generator using SQLite and the Groq API.	The AI recommended a client-server architecture with frontend, Flask backend, AI integration, and database components.	Prepared the Solution Architecture document.
4	Develop the backend.	No specific system prompt/persona used – general conversational query.	Generate Flask code for user authentication, Groq API integration, caption history, and SQLite database connection.	The AI generated modular Flask code with routes, database operations, and API integration.	Implemented the backend successfully.
5	Design the frontend.	No specific system prompt/persona used – general conversational query.	Create responsive HTML and CSS pages for login, registration, dashboard, history, and caption generation.	The AI generated responsive frontend code with improved user interface design.	Developed the frontend pages.
6	Debug the application.	No specific system prompt/persona used – general conversational query.	Help identify and fix Flask routing errors, SQLite issues, template rendering problems, and API integration errors.	The AI suggested debugging steps, corrected routes, fixed database queries, and resolved template errors.	Application executed successfully.

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7	Prepare project documentation.	No specific system prompt/persona used – general conversational query.	Generate project documentation including Problem Statement, Empathy Map, Customer Journey Map, Solution Requirements, Technology Stack, DFD, and Solution Architecture for CaptionAI.	The AI produced structured documentation content tailored to the project.	Documentation prepared and reviewed.
8	Improve project presentation.	No specific system prompt/persona used – general conversational query.	Suggest future enhancements and deployment strategies for CaptionAI.	The AI recommended multilingual support, image-based caption generation, analytics, and cloud deployment.	Future scope and deployment plan documented.

AI Tool Used

Tool Name	ChatGPT
Purpose	Idea generation, code assistance, debugging, documentation writing, architecture planning, and testing support.